Burst-Robust Codes for Life and Qubits

Holographic Erasure Thresholds under Cosmic-Ray Track Structure, and the Neutrino Floor of Repair

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Abstract

Reliability models often begin with independent errors, although ionizing radiation deposits energy in spatially correlated tracks. In superconducting devices, radiation-associated phonon bursts can transiently degrade multiple qubits; in cells, high-linear-energy-transfer (LET) particles produce clustered DNA lesions whose repair differs from that of isolated lesions. We formulate a substrate-agnostic description of burst damage as a marked spatial point process coupled to finite repair capacity. The framework separates geometric code correctability, damage-generation statistics, and repair-service stability. A queue-stability condition, E[S] < , gives a necessary workload criterion for stationary repair; a code-specific erasure criterion separately determines whether a burst footprint is reconstructable. A reproducible benchmark compares iid erasures with finite straight-track bursts on a 24 × 24 lattice at approximately matched marginal site-erasure probability. Across 4,000 Monte Carlo trials per point, track bursts yield a 5.875% spanning-failure frequency at marginal probability 0.40, versus zero observed iid spanning events; at 0.50, the rates are 36.15% and 5.825%. These are finite-size geometric-surrogate results, not logical-error rates of a physical code or measurements of biological damage. Holographic codes motivate recovery geometry, radiobiology supplies the track-structure motivation, and learned decoders suggest an inference strategy. The neutrino floor is treated only as a detector-background analogy, not as a biological mechanism.

1. Motivation and scope

A study published 19 October 2026 reports statistically significant reductions of up to 30.5% in both T1 and T2 immediately following dual detector events attributed to penetrating muons [1]. This is evidence of radiation-associated transient decoherence, not proof that one muon destroys an entire quantum computer. Earlier work also identifies spatially and temporally correlated errors in superconducting devices [2].

High-LET charged particles deposit energy densely along tracks. Their lesions can include clustered double-strand breaks (DSBs), nearby base damage and strand breaks, and chromosomal-scale consequences not captured by independent lesion counts [3–6]. The common abstraction is not that a cell is a quantum computer. It is that both are spatially extended information-bearing systems exposed to correlated disturbances and constrained recovery resources.

Contributions: a marked point-process model for burst footprints; separation of geometric correctability and repair-queue stability; an operational burst-risk functional; a reproducible finite-size benchmark; and a falsifiable cross-domain experimental programme.

2. Common mathematical language

Let G=(V,E) be a finite interaction graph with |V|=n. A damage event is marked by

$$Z_k = (T_k, X_k, \Theta_k, L_k, W_k, M_k),$$

where T is event time, X its anchor, Θ orientation, L extent, W width, and M severity or lesion-complexity mark. It generates footprint C(Z_k) V and repair workload S(Z_k) 0. Events have mean rate ; Poisson arrivals are a

baseline, not a universal assumption. The damaged set is

$$D(t) = \bigcup_{k:T_k \le t < T_k+\tau_k} C(Z_k).$$

A repair controller serves workload at total rate at most . Outstanding workload obeys

$$Q(t+\Delta) \ge [Q(t) + A(t, t+\Delta) - \mu\Delta]_+.$$

The burst profile is the joint law of footprint size, radius, aspect ratio, connected-component structure, and severity, not merely mean damaged-site count.

Proposition 1 — Necessary queue-stability condition

For a stationary ergodic single-server fluid repair queue with workload arrival rate E[S] and constant service rate , a stationary workload distribution with finite mean requires

$$\rho = \frac{\lambda \mathbb{E}[S]}{\mu} < 1.$$

Proof. Over an interval of length t, cumulative arriving workload divided by t converges to E[S]. Service removes at most t. If mean input is at least service capacity, long-run net input is nonnegative; in the non-degenerate single-server setting a finite-mean stationary workload cannot be maintained. This is necessary, not sufficient for every correlated-arrival or priority model.

3. Correctability is not repair

For a quantum code with code projector P and erased subsystem R, exact erasure correction is governed by the Knill–Laflamme condition. In the full-code case, one form is

$$PO_R P = c(O_R)P \quad \text{for all operators } O_R \text{ supported on } R.$$

Holographic codes add geometric structure: bulk operators may be reconstructed from boundary regions whose entanglement wedges contain the relevant bulk support [7–10]. This does not imply that arbitrary radiation tracks are always correctable.

No equivalent exact quantum-code condition is assumed for cells. A biological footprint is mapped to a pathway-dependent workload, residual lesion burden, and functional endpoint:

$$\Phi(C, M; \eta) = \Pr\{\text{residual burden} > \eta \mid C, M, \text{substrate}\}.$$

Quantum decoding, DNA repair, and cellular survival use different Φ functions and observables.

Let F be the family of footprints that are uncorrectable or functionally catastrophic under a specified substrate rule, and q_F = Pr{C(Z) F}. For Poisson event arrivals, thinning gives the exact horizon risk

$$P_{\text{cat}}(T) = 1 - \exp[-\lambda q_F T]. \tag{1}$$

This excludes cumulative sub-catastrophic workload, overlap, repair during the horizon, and non-Poisson clustering. The operational safe region is

$$\mathcal{S} = \{(\mathsf{B}, \lambda, \mu, \mathcal{C}) : \rho < 1,\ P_{\text{cat}}(T) \le \epsilon,\ \Pr(Q(t) > q_{\max}) \le \delta\}. \tag{2}$$

There is no universal scalar threshold independent of geometry, burst law, repair capacity, horizon, and tolerated risk.

4. Holographic recovery and quantum hardware

Holographic quantum error-correcting codes encode bulk degrees of freedom nonlocally across boundary degrees of freedom. Entanglement-wedge reconstruction explains why a bulk operator may remain recoverable after erasure of some boundary subregions, provided the surviving region retains the relevant wedge [7,8]. Code distance, geometry, and erasure location remain decisive; holographic structure is not automatic protection against every burst.

We propose to evaluate a specified code family under

$$\mathcal{N}_{\mathrm{burst}}(\rho) = \sum_{z} p(z)\, \mathrm{Tr}_{C(z)}(\rho) \otimes \sigma_{C(z)}$$

and report logical failure over the full burst-profile distribution, not only an iid-erasure threshold. A hardware experiment should compare matched-rate iid and event-conditioned correlated channels using the same code, decoder, cycle budget, and logical observable.

5. Track-structured biological damage

High-LET heavy ions produce dense ionization patterns and complex DNA damage. Reviews describe clustered DSBs and non-DSB oxidative clustered lesions, with repair kinetics and pathway choice depending on lesion complexity and radiation quality [3–6]. Recent track-structure simulations across H–Fe ions quantify changes in cluster multiplicity with LET and ion species [11]. These findings motivate spatial models; they do not establish that every track overwhelms repair.

The dCas9–KillerRed system offers a controllable method to generate localized ROS-associated DNA damage at targeted genomic regions. Its foundational study reports rapid light-triggered clustered oxidative damage and rapid repair under the tested conditions, with low observed mutagenesis in that setting [12]. A 2025 review surveys genetically encoded photosensitizers and spatially and temporally controlled ROS production, including KillerRed-based DNA-damage tools [17]. It is a calibration and perturbation platform, not proof of a heavy-ion-equivalent track or a general repair-overload phenotype. Candidate measurements include DSB-cluster imaging, repair-factor recruitment time courses, sequencing of repair outcomes, chromosome aberrations, and clonogenic survival. These endpoints must not be conflated.

6. Neural decoding and adaptive repair allocation

AlphaQubit 2 demonstrates high-accuracy, real-time neural decoding for topological codes under the noise settings studied [13]. This motivates training and testing decoders on measured correlated burst channels, but does not establish that AlphaQubit 2 has already been validated on the particular radiation-burst distribution proposed here. Held-out device runs and distribution-shift tests are essential.

A biological repair controller could allocate finite pathway capacity by solving

$$\min_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{H} c(x_t, u_t) + \gamma \ell(x_H)\right], \qquad \sum_{j} u_{j,t} \le \mu.$$

This is a proposed control problem, not a demonstrated intervention. Observability, feasible actions, safety constraints, and reward validity must be established before reinforcement learning is used.

7. Synthetic benchmark

We implement a finite-size geometric surrogate on a 24×24 square lattice. Failure is a 4-neighbour connected erased component spanning horizontally or vertically. In the iid channel, each site is erased independently with probability p. In the burst channel, a Poisson number of finite five-site line tracks are placed with horizontal, vertical, and diagonal orientations; intensity is chosen to approximately match the same marginal site-erasure probability. Each point uses 4,000 trials and fixed seed 20261106.

p	iid failures	track failures	Difference
0.30	0/4000	11/4000	0.275 pp
0.35	0/4000	61/4000	1.525 pp
0.40	0/4000	235/4000	5.875 pp
0.45	18/4000	658/4000	16.000 pp
0.50	233/4000	1446/4000	30.325 pp
0.55	1251/4000	2513/4000	31.525 pp
0.60	2998/4000	3331/4000	8.325 pp

At p = 0.40, zero iid events in 4,000 trials does not mean zero probability; the rule-of-three gives an approximate 95% upper bound of 3/4000 = 0.075%. The burst frequency is 5.875%. At p = 0.50, estimates are 5.825% and 36.15%. The curves approach one another at high density; the track channel is not uniformly worse at every point.

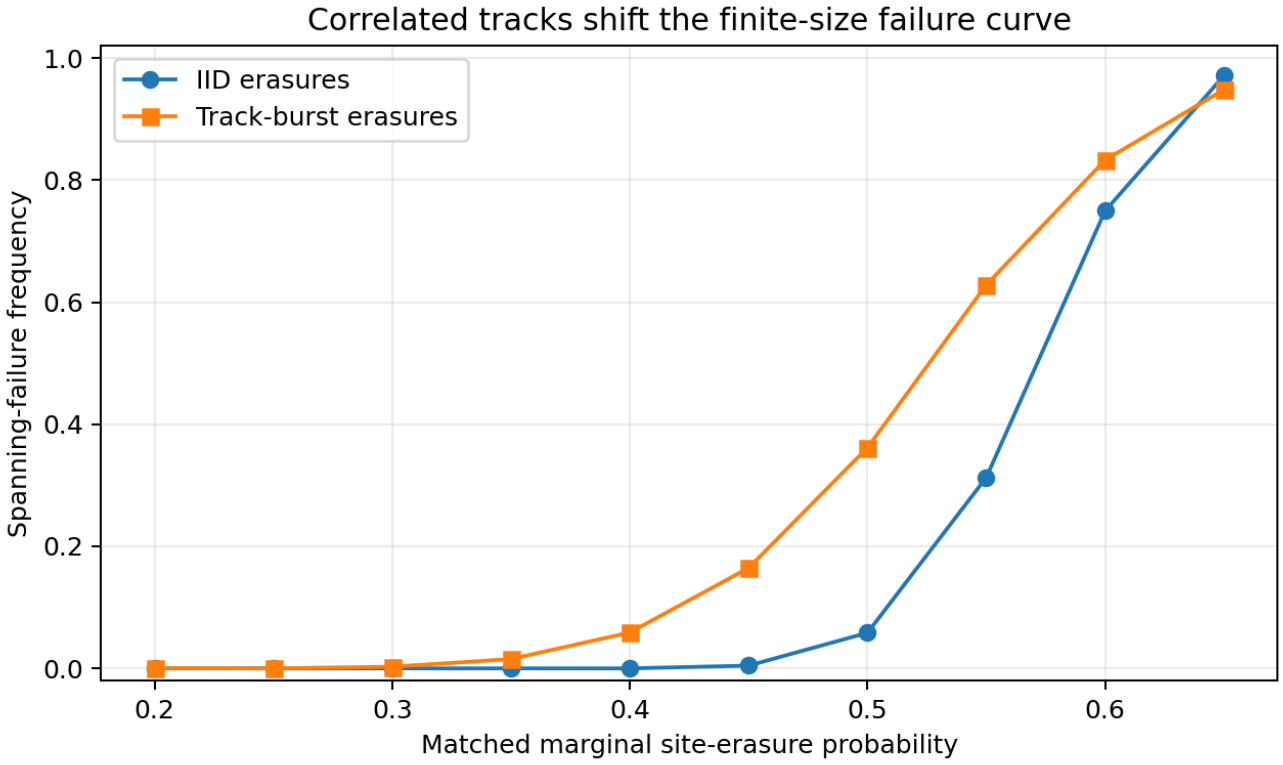


Figure 1: Finite-size spanning-failure frequencies

The benchmark shows that spatial correlation changes a geometric connectivity statistic at approximately matched marginal occupancy. It does not simulate a surface-code syndrome, holographic tensor network, quantum coherence, DNA chemistry, chromatin, or repair enzyme. Its values must not be called a quantum erasure threshold or cellular damage threshold.

8. Quantum biology and the neutrino boundary

Engineered fluorescent proteins have demonstrated optically detected magnetic resonance in living bacterial cells at room temperature, providing a biological quantum-sensing platform [14]. This is a quantum sensor based on spin-dependent photophysics; it does not make a cell a fault-tolerant quantum computer, nor does it demonstrate proton tunnelling as a cause of organism-wide mutation.

The "neutrino floor" has a precise meaning in dark-matter direct detection: coherent elastic neutrino–nucleus scattering creates a difficult-to-disentangle background for certain searches [15,16]. It is not a biological repair

floor. Solar neutrinos are not proposed as a meaningful cellular damage source. The phrase in the title is an analogy for an inference limit in a separate detector domain, not a physical mechanism.

9. Limitations and falsifiable programme

This is a theory-plus-synthetic-benchmark preprint. It does not establish a universal burst-erasure law, a new holographic code, a biological repair mechanism, or measured improvements in quantum or cellular survival. The lattice generator should be replaced by calibrated event distributions. A holographic-code study needs a specified code, erasure channel, decoder, and finite-size scaling. A biological study needs radiation-quality-specific track data and preregistered repair and survival endpoints.

1. **Quantum:** collect event-conditioned qubit error maps; fit spatial-temporal burst kernels; compare surface, colour, and holographic codes at matched marginal rates and equal decoding latency.
2. **Radiobiology:** fit track kernels from transport simulations and microscopy; calibrate workload marks against lesion complexity; compare matched-total-dose burst and dispersed exposures.
3. **Repair control:** measure pathway-specific service distributions and test queue predictions before learning a repair policy.
4. **Cross-domain:** compare dimensionless burst-size/code-distance, workload/capacity, and persistence/recovery-time ratios, not biological and quantum observables as if identical.

10. Conclusion

A common mathematical language can describe spatial bursts, recoverability, and finite repair without asserting a common microscopic mechanism. Average error rate alone is insufficient: burst profile and recovery capacity must be measured jointly. Holographic codes offer a geometric vocabulary for quantum reconstruction; radiation track structure motivates clustered biological damage; queueing theory supplies a workload constraint; and learned decoders offer an inference strategy. The synthetic benchmark is a deliberately limited proof of concept. The defensible contribution is a testable framework and experimental agenda, not the claim that one muon defeats every quantum computer or one heavy ion necessarily defeats every cell.

Reproducibility statement

All numerical values are generated from the included deterministic synthetic model. No human, animal, patient, or proprietary quantum-hardware data were used. A passing code check is not empirical validation or peer review.

References

1. A. R. Castelli et al., "Superconducting-qubit decoherence correlated with detected radiation events," *Physical Review Applied* 26, 044052 (2026), DOI: 10.1103/ddyy-t37l.
2. H. Binney et al., "Distinguishing types of correlated errors in superconducting qubits," *Physical Review Applied* 26, 024025 (2026), DOI: 10.1103/j235-47t6.
3. Y. Hagiwara, T. Oike, and A. Niimi, "Clustered DNA double-strand break formation and the repair pathway following heavy-ion irradiation," *Journal of Radiation Research* 60, 69–79 (2019), DOI: 10.1093/jrr/rry096.
4. A. Shibata, "Carbon ion radiation and clustered DNA double-strand breaks," *The Enzymes* 51, 117–130 (2022), DOI: 10.1016/bs.enz.2022.08.008.
5. D. T. Goodhead and M. Weinfeld, "Clustered DNA Damage and its Complexity: Tracking the History," *Radiation Research* 202, 385–407 (2024), DOI: 10.1667/RADE-24-00017.1.
6. B. Wilkinson, M. A. Hill, and J. L. Parsons, "The Cellular Response to Complex DNA Damage Induced by Ionising Radiation," *International Journal of Molecular Sciences* 24, 4920 (2023), DOI: 10.3390/ijms24054920.
7. F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, "Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence," *JHEP* 06, 149 (2015), DOI: 10.1007/JHEP06(2015)149.
8. D. Harlow, "The Ryu–Takayanagi Formula from Quantum Error Correction," *Communications in Mathematical Physics* 354, 865–912 (2017), DOI: 10.1007/s00220-017-2904-z.
9. R. J. Harris et al., "Calderbank-Shor-Steane holographic quantum error-correcting codes," *Physical Review A* 98, 052301 (2018), DOI: 10.1103/PhysRevA.98.052301.

10. N. Bao, C. Cao, and G. Zhu, "Deconfinement and error thresholds in holography," *Physical Review D* 106, 046009 (2022), DOI: 10.1103/PhysRevD.106.046009.
11. "Comprehensive database of track-structure simulations on DNA damage by H–Fe ions up to 1 GeV/u for space radiation biology," *Scientific Reports* (2026), DOI: 10.1038/s41598-026-55264-8.
12. J. Wang et al., "Site-specific targeting of a light activated dCas9-KillerRed fusion protein generates transient, localized regions of oxidative DNA damage," *PLOS ONE* 15, e0237759 (2020), DOI: 10.1371/journal.pone.0237759.
13. A. W. Senior et al., "A scalable and real-time neural decoder for topological quantum codes," arXiv:2512.07737 (2025).
14. G. Abrahams et al., "Quantum spin resonance in engineered proteins for multimodal sensing," *Nature* (2026).
15. E. Aprile et al. (XENON Collaboration), "Probing the Solar 8B Neutrino Fog with XENONnT," *Physical Review Letters* 137, 091807 (2026), DOI: 10.1103/xvpn-w3q9.
16. Z. Tabrizi et al., "Clarity through the neutrino fog: constraining new forces in dark matter detectors," *JHEP* (2025), DOI: 10.1007/JHEP08(2025)043.
17. A. Dey and R. P. Barnes, "Genetic approaches for targeted oxidative stress," *NAR Cancer* 7(4), zcaf049 (2025), DOI: 10.1093/narcan/zcaf049.